MODULE 1 · LECTURE 18

Preparation of Crop Calendars Using IT Tools

What crop calendars are and why farmers, extension staff and planners use them; how operations are timed by days after sowing, growth stages and thermal time; and how spreadsheets, programs, calendar files, apps and FAO's online tool turn a plan into dated reminders.

Learning objectives

After this lecture you should be able to:

1. Define a crop calendar and describe its types and uses.
2. Describe India's cropping seasons and typical sowing and harvest windows.
3. Time operations by days after sowing, growth stages (Zadoks scale) and growing degree days.
4. Prepare a crop calendar in a spreadsheet, including date arithmetic and a Gantt chart.
5. Generate a dated calendar and an iCalendar file with a short program.
6. Use online crop-calendar tools and datasets, and adjust calendars for weather.

Scope: this lecture brings together tools from earlier lectures — spreadsheets (Lecture 5), programming (Lecture 8), growing degree days and agromet advisories (Lecture 11), crop water requirement and crop stages (Lecture 13), apps (Lecture 14), maps (Lecture 15) and decision support (Lecture 16). Artificial intelligence begins in Lecture 19.

1. What is a crop calendar?

A **crop calendar** is a schedule of the sowing and harvesting periods of crops and of the operations carried out between them — land preparation, sowing, fertiliser application, irrigation, weeding, plant protection and harvest — for a given crop, region and season. A good calendar tells a farmer **what to do, when, and how much**, and helps everyone along the supply chain plan inputs, labour, machinery, water, credit and marketing.

Type	What it shows	Main users
Crop operation calendar	Dated operations for one crop from field preparation to harvest	Farmers, extension workers
Regional crop calendar	Sowing and harvest windows of the main crops of a region	Planners, input suppliers, crop-monitoring agencies
Cropping-system calendar	The sequence of crops through the year on a field	Farm planners
Pest and disease calendar	Periods of risk for major pests and diseases by crop stage	Plant-protection staff, farmers
Input and labour calendar	When seed, fertiliser, water, labour and machines are needed	FPOs, dealers, custom-hiring centres

2. Cropping seasons in India

- **Kharif** — crops sown with the south-west monsoon (about June–July) and harvested in autumn: rice, maize, sorghum, pearl millet, soybean, groundnut, cotton.

- **Rabi** — crops sown after the monsoon (about October–December) and harvested in spring: wheat, mustard, chickpea, barley.
- **Zaid** — short crops grown between rabi and kharif (about March–June), usually with irrigation: green gram, vegetables, fodder.

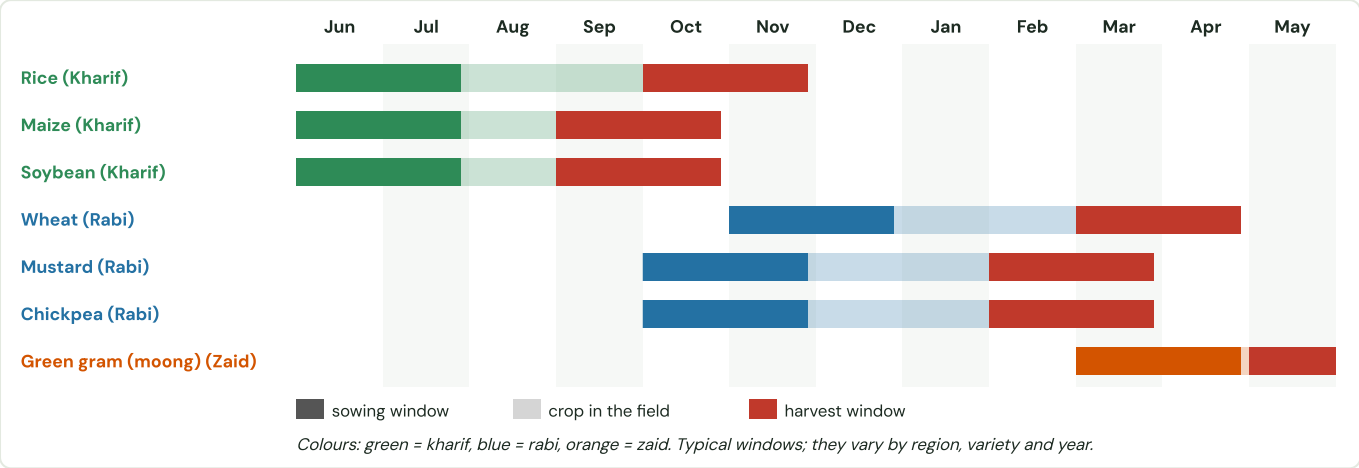


Figure 1. A regional crop calendar showing typical sowing and harvest windows (illustrative; actual windows vary by region, variety and year). Original chart.

3. Timing operations

Operations can be timed in three ways, from simplest to most precise:

Method	Basis	Strength	Weakness
Days after sowing (DAS)	Fixed number of days from sowing	Simple; easy to compute and remember	Ignores weather: crops develop faster in warm years
Growth stages	Visible development of the crop	Matches the crop's actual state	Needs field observation
Thermal time (GDD)	Accumulated heat above a base temperature (Lecture 11)	Predicts stages ahead of time and adjusts for weather	Needs temperature data and crop-specific thresholds

Growth-stage scales

The **Zadoks decimal code** describes cereal development with two digits: the first gives the principal stage and the second the detail within it (Zadoks et al., 1974). The ten principal stages are 0 germination, 1 seedling growth, 2 tillering, 3 stem elongation (jointing), 4 booting, 5 ear emergence, 6 flowering (anthesis), 7 milk development, 8 dough development and 9 ripening. Using stage names rather than dates makes advice valid whatever the sowing date.

Critical stages for irrigation in wheat

Wheat responds most to irrigation at **crown root initiation (CRI), tillering, jointing, flowering, milk and dough** stages; in a field experiment in Karnataka, irrigation at all six stages gave the most effective tillers and dry matter (Ahmad & Kumar, 2015). When water is limited, irrigations are concentrated on the most sensitive of these stages, beginning with CRI.

The days after sowing used for these stages in the examples below are approximate values commonly quoted for timely-sown wheat; actual timing depends on variety, sowing date, soil and weather, so the local package of practices should always be followed.

Adjusting for weather with growing degree days

Worked example 1 — predicting flowering by thermal time (illustrative temperatures and threshold)
Wheat sown on 15 November; assume flowering after 1,300 degree-days above a base of 0 °C. Mean temperatures: November 20 °C, December 15 °C, January 13 °C, February 16 °C.
Normal year: 15–30 November = 16 days × 20 = 320; December = 31 × 15 = 465 (total 785); January = 31 × 13 = 403 (total 1,188). Remaining 112 ÷ 16 = 7 days → flowering about **7 February** (84 DAS).
A year 2 °C warmer: 16 × 22 = 352; 31 × 17 = 527 (total 879); remaining 421 ÷ 15 = 28.1 days → about **29 January** (75 DAS) — 9 days earlier.
A calendar fixed in days would schedule the flowering irrigation too late in the warm year; a GDD-based calendar moves it automatically.

4. Preparing a crop calendar with IT tools

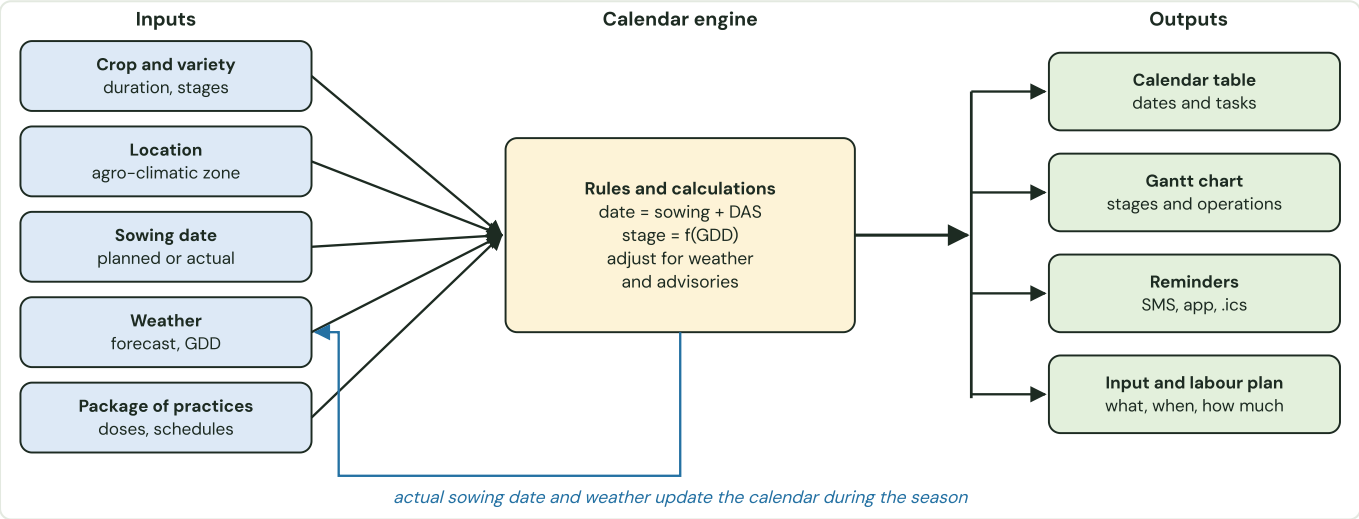


Figure 2. How an IT-based crop-calendar tool works. Original diagram.

4.1 Spreadsheets

Spreadsheets store dates as numbers of days, so a date plus a number of days is another date (Lecture 5). A simple calendar sheet has one row per operation:

Column	Content	Formula (row 2)
A	Operation	typed: Irrigation at CRI
B	Days after sowing	typed: 21
C	Date	=$F$1 + B2 (sowing date entered once in F1)
D	Day of the week	=TEXT(C2, "ddd")
E	Days from today	=C2 - TODAY()
G2 onwards	Gantt chart cell for each week	conditional format: =AND(G$1 >= $C2, G$1 < $C2 + 7)

Worked example 2 — spreadsheet date arithmetic
Sowing date in F1 = 15-Nov-2026. Operation "Irrigation at jointing" has B = 65 → C = 15-Nov-2026 + 65 = 19-Jan-2027; D shows "Tue".
If sowing is delayed to 25 November, changing only F1 moves every date in column C by 10 days — the main advantage of a formula-based calendar.

4.2 A short program

A few lines of Python (Lecture 8) produce the same calendar and export it as an **iCalendar (.ics)** file, the standard format that phone and computer calendar applications can import (Desruisseaux, 2009). Each operation becomes an event with a reminder on the right day.

Generating a wheat calendar and an .ics file in Python

```
from datetime import date, timedelta

SOWING = date(2026, 11, 15)
PLAN = [  # (operation, days after sowing) - approximate; follow the local package of practices
    ("Field preparation and pre-sowing irrigation", -10),
    ("Sowing; basal fertiliser", 0),
    ("Irrigation at crown root initiation; top-dress nitrogen", 21),
    ("Weed control", 30),
    ("Irrigation at tillering", 45),
    ("Irrigation at jointing", 65),
    ("Irrigation at flowering", 85),
    ("Irrigation at milk stage", 105),
    ("Harvest", 120),
]

for task, das in PLAN:
    print(f"{SOWING + timedelta(days=das):%d %b %Y}  ({das:+4d} DAS)  {task}")

# Export the same plan as an iCalendar (.ics) file for phone or computer calendars
lines = ["BEGIN:VCALENDAR", "VERSION:2.0", "PRODID:-//VAC-II//Crop calendar//EN"]
for i, (task, das) in enumerate(PLAN):
    day = SOWING + timedelta(days=das)
    lines += ["BEGIN:VEVENT", f"UID:wheat-2026-{i}@example.org", "DTSTAMP:20261109T000000Z",
              f"DTSTART;VALUE=DATE:{day:%Y%m%d}", f"SUMMARY:Wheat - {task}", "END:VEVENT"]
lines.append("END:VCALENDAR")
with open("wheat_calendar.ics", "w", newline="") as f:
    f.write("\r\n".join(lines) + "\r\n")
print(len(PLAN), "events written to wheat_calendar.ics")

Output:
05 Nov 2026  ( -10 DAS)  Field preparation and pre-sowing irrigation
15 Nov 2026  (  +0 DAS)  Sowing; basal fertiliser
06 Dec 2026  ( +21 DAS)  Irrigation at crown root initiation; top-dress nitrogen
15 Dec 2026  ( +30 DAS)  Weed control
30 Dec 2026  ( +45 DAS)  Irrigation at tillering
19 Jan 2027  ( +65 DAS)  Irrigation at jointing
08 Feb 2027  ( +85 DAS)  Irrigation at flowering
28 Feb 2027  (+105 DAS)  Irrigation at milk stage
15 Mar 2027  (+120 DAS)  Harvest
9 events written to wheat_calendar.ics
```

The program was run and the output shown is its actual output; the .ics file it wrote contains nine events. The days after sowing are approximate (see Section 3).

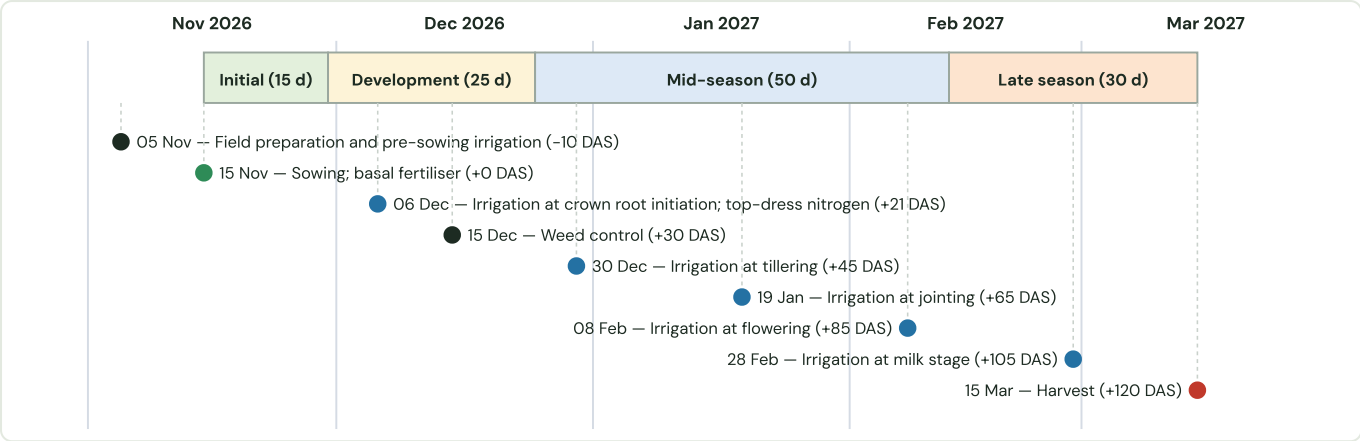


Figure 3. Wheat crop calendar produced from the program, with FAO-56 growth stages for wheat sown in November in central India (initial 15, development 25, mid-season 50 and late season 30 days). Original chart.

4.3 Online tools, data and apps

Tool or data	What it offers
FAO Crop Calendar	A free web application giving planting, sowing and harvesting periods for over 100 crops by country and agro-ecological zone, kept up to date with data validated by national authorities (Food and Agriculture Organization of the United Nations [FAO], n.d.)
Global crop-calendar datasets	Sacks et al. (2010) assembled planting and harvesting dates for 19 major crops worldwide from six sources and related them to climate — used in crop models and monitoring
FAO-56 stage lengths	Typical lengths of crop growth stages by region and sowing month, used to build stage-based calendars and crop-water schedules (Allen et al., 1998; Lecture 13)
Agromet advisories and apps	Weekly district advisories (every Tuesday and Friday) and apps such as Meghdoot adjust operations to the weather forecast (Lectures 11 and 14)
Calendar apps and SMS	Reminders from .ics files, app notifications or text messages
GIS	Regional calendars mapped by district or agro-climatic zone (Lecture 15)

4.4 Steps in preparing a crop calendar

1. Choose the crop, variety (its duration) and location; obtain the recommended sowing window.
2. List the operations and the recommended doses from the local package of practices.
3. Fix each operation by DAS, growth stage or GDD threshold.
4. Enter the sowing date and compute dates in a spreadsheet or program; draw a Gantt chart.
5. Add input quantities, labour and machinery needs for each operation.
6. Share the calendar as a table, chart, .ics file or app reminders.
7. Update it during the season with the actual sowing date, observed growth stages, weather forecasts and advisories.

5. Benefits and limitations

Benefits	Limitations
Timely operations and fewer missed critical stages	Fixed-date calendars ignore weather variation
Advance planning of inputs, labour, machinery and credit	Generic calendars may not fit local varieties and soils
Easy updating when sowing is delayed	Need reliable sowing dates, temperature data and stage thresholds
Reminders reach farmers through phones	Farmers still need field observation to confirm stages
Regional calendars support crop monitoring and policy	Calendars must be revised as varieties and climate change

6. Key takeaways

- A crop calendar schedules sowing, harvest and the operations between them for a crop, region and season.
- Calendars may be for one crop, a region, a cropping system, pests or inputs and labour.
- India's seasons are kharif (monsoon-sown), rabi (post-monsoon) and zaid (summer, between them).
- Operations are timed by days after sowing, growth stages (Zadoks scale) or growing degree days; GDD adjusts timing to the weather.
- Wheat's critical irrigation stages are CRI, tillering, jointing, flowering, milk and dough.
- Spreadsheets (date + DAS, Gantt via conditional formatting) and short programs generate calendars; iCalendar (.ics) files deliver reminders.
- FAO's Crop Calendar, global planting-date datasets, FAO-56 stage lengths, agromet advisories and apps support calendar preparation.

7. Examination questions

Short-answer questions (2 marks)

Q1. Define a crop calendar.

A schedule of the sowing and harvest periods of crops and of the operations between them for a crop, region and season.

Q2. Name three types of crop calendar.

Crop operation calendar, regional crop calendar and cropping-system calendar (also pest and input calendars).

Q3. Distinguish kharif, rabi and zaid seasons.

Kharif: sown with the monsoon, harvested in autumn; rabi: sown after the monsoon, harvested in spring; zaid: short summer crops between them.

Q4. What does DAS stand for, and what is its weakness?

Days after sowing; it ignores weather, so stages may come earlier or later than planned.

Q5. What is the Zadoks scale?

A two-digit decimal code for the growth stages of cereals, from 0 (germination) to 9 (ripening).

Q6. List the critical irrigation stages of wheat.

Crown root initiation, tillering, jointing, flowering, milk and dough.

Q7. Write a spreadsheet formula for the date of an operation 21 days after a sowing date in cell F1.

=$F$1 + 21 (or =$F$1 + B2, with 21 in B2).

Q8. What is an iCalendar (.ics) file?

A standard calendar file that calendar applications can import as dated events.

Q9. What does the FAO Crop Calendar provide?

Planting, sowing and harvesting periods for over 100 crops by country and agro-ecological zone.

Q10. Why are GDD-based calendars better than fixed-date calendars?

They adjust the timing of stages and operations to the actual temperatures of the season.

Long-answer questions (5–10 marks) — answer outlines

Q1. Explain the types and uses of crop calendars.

Definition; types table; uses for farmers, suppliers, planners.

Q2. Describe India's cropping seasons with a regional crop calendar.

Kharif, rabi, zaid with crops; Figure 1.

Q3. Compare methods of timing farm operations and explain growth-stage scales.

DAS, growth stages, GDD (table); Zadoks scale; wheat critical stages; worked example 1.

Q4. Explain how a crop calendar is prepared in a spreadsheet.

Column layout and formulas; worked example 2; Gantt chart by conditional formatting.

Q5. Describe how a program can generate a crop calendar and reminders.

Python listing and output; iCalendar file; Figure 3.

Q6. Discuss IT tools and data sources for crop calendars and how calendars are kept up to date.

Figure 2; FAO Crop Calendar, Sacks et al. data, FAO-56, advisories and apps, GIS; steps; updating with weather.

8. Glossary

Term	Meaning
Anthesis	Flowering — the release of pollen
Crop calendar	Schedule of crop sowing, harvest and operations
CRI	Crown root initiation — early wheat stage critical for irrigation
DAS	Days after sowing
Gantt chart	Bar chart of tasks against time
GDD	Growing degree days — accumulated heat above a base temperature
iCalendar (.ics)	Standard file format for calendar events
Kharif	Monsoon-sown cropping season
Package of practices	Official recommendations for growing a crop in a region
Rabi	Post-monsoon (winter) cropping season
Sowing window	Recommended period for sowing a crop
Zadoks scale	Decimal code for cereal growth stages
Zaid	Short summer cropping season between rabi and kharif

References

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Desruisseaux, B. (Ed.). (2009). *Internet calendaring and scheduling core object specification (iCalendar)* (RFC 5545). RFC Editor. https://doi.org/10.17487/RFC5545

Food and Agriculture Organization of the United Nations. (n.d.). *Crop Calendar* [AgriTech Observatory entry]. Retrieved November 9, 2026, from https://agritechobservatory.review.fao.org/crop-calendar

Sacks, W. J., Deryng, D., Foley, J. A., & Ramankutty, N. (2010). Crop planting dates: An analysis of global patterns. *Global Ecology and Biogeography, 19*(5), 607–620. https://doi.org/10.1111/j.1466-8238.2010.00551.x

Zadoks, J. C., Chang, T. T., & Konzak, C. F. (1974). A decimal code for the growth stages of cereals. *Weed Research, 14*(6), 415–421. https://doi.org/10.1111/j.1365-3180.1974.tb01084.x

All diagrams, charts and the program are original. Sowing and harvest windows, days after sowing, temperatures and the GDD threshold in the examples are illustrative; FAO-56 stage lengths are from the cited source.

Practice MCQs

Q1. A crop calendar mainly shows

(a) crop prices
(b) the timing of sowing, operations and harvest
(c) soil types
(d) farm boundaries

Q2. Wheat, mustard and chickpea in India are typically

(a) kharif crops
(b) rabi crops
(c) zaid crops
(d) perennial crops

Q3. Crops sown with the south-west monsoon belong to the

(a) rabi season
(b) zaid season
(c) kharif season
(d) winter season

Q4. The main weakness of timing operations by days after sowing is that it

(a) is hard to calculate
(b) ignores weather variation
(c) needs satellites
(d) cannot be put in a spreadsheet

Q5. In the Zadoks scale, principal stage 6 is

(a) tillering
(b) booting
(c) flowering (anthesis)
(d) ripening

Q6. The first critical irrigation stage of wheat is

(a) dough
(b) crown root initiation
(c) milk
(d) booting

Q7. If the sowing date is in F1 and the DAS in B2, the operation date is

(a) =F1 - B2
(b) =$F$1 + B2
(c) =B2 ÷ F1
(d) =TODAY()

Q8. A warmer-than-normal season makes a GDD-based calendar schedule flowering

(a) later
(b) earlier
(c) at the same date
(d) not at all

Q9. The standard file format for exporting events to calendar apps is

(a) .csv
(b) .ics (iCalendar)
(c) .shp
(d) .pdf

Q10. The FAO Crop Calendar provides

(a) mandi prices
(b) planting, sowing and harvesting periods by crop and zone
(c) soil-test values
(d) weather radar

Q11. Sacks et al. (2010) compiled global planting and harvesting dates for

(a) 5 crops
(b) 19 major crops
(c) 100 crops
(d) 500 crops

Q12. A Gantt chart shows

(a) tasks against time
(b) yield against rainfall
(c) price against quantity
(d) soil depth

Answers

Q1 — (b) the timing of sowing, operations and harvest
Q2 — (b) rabi crops
Q3 — (c) kharif season
Q4 — (b) ignores weather variation
Q5 — (c) flowering (anthesis)
Q6 — (b) crown root initiation
Q7 — (b) =$F$1 + B2
Q8 — (b) earlier
Q9 — (b) .ics (iCalendar)
Q10 — (b) planting, sowing and harvesting periods by crop and zone
Q11 — (b) 19 major crops
Q12 — (a) tasks against time